

RCET 2253

Clinical Assistant Professor

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Discrete Linear Regulated Power Supply Project

Check-Off Sheet

Name:	Start Date:	
Check-Offs	Date	Instr. Int.
1. Calculated rectifier section with predicted waveforms.		
2a. Assemble/Solder: power cord, fuses, on/off switch, transformer, & rectifier. Perform continuity checks.		
2b. Measure and document rectified unloaded voltage waveforms.		
2c. Measure and document loaded voltage waveforms.		
3a. Calculate appropriate filter cap & On/Off indication circuit.		
3b. Assemble Filter Capacitor and measure unloaded and loaded voltage waveforms.		
4a. Calculate and document the op-amp regulated variable voltage section.		
4b. Assemble/solder op-amp regulated variable voltage section. With no load (RL=10K), verify variable voltage operation.		
5a. Calculate and document the current- limiting section		
5b. Assemble/solder the current-limiting section. Verify current-limiting operation by incrementally increase load until output is shorted through your current meter. Verify limiting range.		
6a. Calculate and document over-voltage protection circuit		
6b. Assemble/solder over-voltage section - verify operation.		
7a. Calculate overvoltage/output-shortcd indication circuit.		
7b. Assemble/solder overvoltage/output- shortcd indication circuit. Verify operation.		
8a. Final Check-Off: Verify variable voltage, current limiting, LED Indicators, and over-voltage.		
8b. Final Check-Off: 60min output shorted test.		
8c. Final Check-Off: Verify variable voltage, current limiting, LED Indicators, and over-voltage.		